SOFTWARE REQUIREMENTS SPECIFICATION

Property Insurance Management System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the Property Insurance Management System (PIMS). It is intended for developers, testers, project managers, and stakeholders involved in the design, development, and deployment of the system.

1.2 Scope

PIMS is a full-stack enterprise web application covering the complete lifecycle of property insurance — from property registration and underwriting through premium calculation, policy issuance, payment processing, claim handling, referral rewards, and settlement.

- Property registration and category management
- Underwriting, risk assessment, and premium calculation
- Policy issuance, activation, and lifecycle tracking
- Simple payment — Amount entry, card details, and one-click Pay Now
- Claim submission, review, and settlement
- Referral Rewards System
- Filter-based search for Admin user and policy management

1.3 Definitions, Acronyms & Abbreviations

Acronym	Definition
PIMS	Property Insurance Management System
SRS	Software Requirements Specification
JWT	JSON Web Token
API	Application Programming Interface
EF Core	Entity Framework Core
RBAC	Role-Based Access Control
FR	Functional Requirement
NFR	Non-Functional Requirement

DTO	Data Transfer Object
DI	Dependency Injection

1.4 Technology Stack

Component	Technology
Backend	ASP.NET Core (.NET 8) Web API
Frontend	Angular (Latest Version)
Database	SQL Server
ORM	Entity Framework Core
Security	JWT Bearer Authentication
Architecture	Clean Architecture — API / Application / Domain / Infrastructure

2. Overall Description

2.1 Problem Statement

The system addresses the following existing problems in manual property insurance management:

- Manual policy processing leading to delays and errors
- Decentralized property data with poor accessibility
- Delayed claim settlements due to lack of automated workflows
- Poor tracking of the policy lifecycle
- Lack of transparency in premium calculation
- No incentive mechanism for customer acquisition

2.2 Proposed Solution

- End-to-end policy lifecycle management with automated workflows
- Structured property category and subcategory management
- Automated premium calculation using a defined formula
- Centralized claim processing with status tracking
- Role-based secured system with JWT
- Referral Rewards System for customer acquisition
- Filter-based admin search for efficient user and policy management
- Simple payment screen for premium collection — amount, card details, and Pay Now

2.3 User Classes

The system serves four distinct user roles: Admin, Agent, Customer, and Claims Officer. Full details are in Section 5.

3. Key Highlights

The following key features distinguish PIMS from conventional insurance platforms:

✓ **Clean Architecture** Strict layered separation: API / Application / Domain / Infrastructure	✓ **Dynamic Fraud Validations** Real-time detection of suspicious claims, policies, and user activity	✓ **Referral Reward System** Auto-credit ₹50 on every referred user's first policy activation
✓ **Light / Dark Mode UI** User-selectable themes for improved accessibility and comfort	✓ **Real-time Notifications** Event-driven alerts for policy, payment, claim, and reward events	✓ **Enterprise-Level Design** JWT, RBAC, exception middleware, structured logging, and pagination

4. Referral Rewards System

The Referral Rewards System enables existing customers to earn reward points by referring new users. The system operates automatically through the four-step workflow below:

Figure 2 — Referral Rewards System Workflow

4.1 Workflow Steps

Step 01	**Customer Signs Up**	New user registers using a referral code from an existing customer. Code is unique per registered customer, auto-generated on account creation.
Step 02	**Referral Detected**	System automatically links the new account to the referrer. No manual action required from the referrer or admin.
Step 03	**Reward Credited**	When the referred user activates their first policy, ₹50 bonus reward points are automatically credited to the referrer. Fully system-detected.
Step 04	**Redeem Rewards**	Accumulated reward points are redeemable instantly for premium discounts or cashback on existing policies.

4.2 Referral Parameters

Parameter	Value / Rule
Reward per Referral	₹50 credited upon referred user's first policy activation
Referral Limit	Unlimited — no cap on referrals per customer
Detection Method	Automated — system links referral code at registration
Credit Timing	Instant — triggered automatically on first policy activation
Redemption Options	Premium discounts or cashback on existing policies

5. User Roles & Responsibilities

5.1 Admin

- Create and manage property categories and subcategories
- Approve or reject insurance policies
- Monitor claims and settlements
- Filter-based search across users and policies (see Section 8.3)
- View reports and system analytics
- Manage users and assign roles

5.2 Agent

- Assist customers in policy creation
- Perform underwriting and risk assessment
- Submit property details on behalf of customers
- Track policy requests and status

5.3 Customer

- Register and login to the system
- Submit property information for insurance
- View policy details and coverage
- Make premium payments by entering amount and card details on the payment screen
- Raise and track insurance claims
- Earn and redeem referral reward points

5.4 Claims Officer

- Review submitted insurance claims
- Verify supporting documents
- Approve or reject claims based on policy validity
- Process claims settlements and generates settlement records

6. Property Insurance Categories & Subcategories

6.1 Residential Property

Types: House Insurance, Apartment Insurance, Villa Insurance, Rental Property Insurance
Coverage: Fire Damage, Natural Disasters, Theft & Burglary, Structural Damage

6.2 Commercial Property

Types: Office Buildings, Shops & Retail Spaces, Warehouses, Factories
Coverage: Fire & Explosion, Machinery Breakdown, Business Interruption, Theft

6.3 Industrial & Special Use

Types: Plant & Machinery, Agricultural Property, Infrastructure, Construction Sites
Coverage: Equipment Failure, Industrial Fire, Structural Collapse, Environmental Risk

6.4 Property Contents

Types: Furniture, Electronics, Equipment, Inventory & Stock
Coverage: Theft, Fire Damage, Accidental Damage, Electrical Surge

7. Functional Requirements

7.1 Authentication & Authorization

- FR-01: Users shall register with name, email, password, and role
- FR-02: System shall authenticate users and issue JWT Bearer tokens
- FR-03: RBAC shall restrict access per role across all API endpoints
- FR-04: Passwords shall be hashed using bcrypt before storage

7.2 Category & Property Management

- FR-05: Admin shall create, update, and delete property categories and subcategories

- FR-06: Customers and Agents shall submit property details for registration
- FR-07: System shall store address, property value, construction type, coverage type, and documents

7.3 Policy Management

- FR-08: System shall create a Draft Policy upon property submission
- FR-09: Premium shall be calculated as: Premium = Property Value × Risk Factor × Coverage Rate
- FR-10: Admin shall approve or reject policy requests
- FR-11: System shall generate Policy Number, coverage details, validity period, and invoice on approval
- FR-12: Policy lifecycle: Draft → Pending Approval → Active → Expired / Cancelled
- FR-13: System shall support policy renewal, cancellation, and expiry tracking with notifications

7.4 Payment

- FR-14: Customer shall enter the premium Amount and Card Details on the payment screen
- FR-15: Customer shall click "Pay Now" to submit the payment
- FR-16: On successful payment, the system shall automatically generate a Payment Invoice
- FR-17: Invoice shall include Policy Number, Amount Paid, Payment Date, and Transaction Reference
- FR-18: Policy shall be activated immediately after invoice is generated
- FR-19: Payment record shall be stored with Amount, Card (last 4 digits), Date, and Status

7.5 Claim Management

- FR-20: Customers shall submit claims with policy number, damage details, and evidence documents
- FR-21: Claims Officer shall review and verify claims
- FR-22: Claim lifecycle: Submitted → Under Review → Approved / Rejected
- FR-23: Settlement Amount = Loss Amount − Deductible
- FR-24: Settled claims shall be recorded with final status 'Settled'

7.6 Referral Rewards System

- FR-25: System shall auto-generate a unique referral code per customer
- FR-26: System shall link referral code at registration automatically
- FR-27: System shall credit ₹50 reward points on referred user's first policy activation
- FR-28: Reward points shall be redeemable for premium discounts or cashback
- FR-29: No limit on number of referrals per customer

7.7 Admin Filter-Based Search

- FR-30: Admin shall filter and search Users with the criteria defined in Section 8.3
- FR-31: Admin shall filter and search Policies with the criteria defined in Section 8.3
- FR-32: All filter fields shall support partial-match / range queries
- FR-33: Filter results shall be paginated and sortable

7.8 Notifications

- FR-34: System shall send real-time notifications for policy approvals, premium due dates, claim status changes, payment results, and reward credits

8. Feature Specifications

8.1 Payment

The payment screen provides a simple, straightforward premium collection flow. The customer enters the payment amount and card details, clicks Pay Now, and the system immediately generates a Payment Invoice and activates the policy.

Field	Description
Amount	Pre-filled with the policy premium amount — editable if partial payment is allowed
Card Number	16-digit card number entered by the customer
Cardholder Name	Name as printed on the card
Expiry Date	Card expiry in MM/YY format
CVV	3-digit security code on the back of the card
Pay Now Button	Submits the payment and triggers invoice generation

Payment Flow

Step 1	**Policy Approved**	Admin approves policy → Customer is notified with the premium amount due
Step 2	**Open Payment Screen**	Customer opens the payment screen — Amount is pre-filled from the approved policy
Step 3	**Enter Card Details**	Customer enters Card Number, Cardholder Name, Expiry Date, and CVV
Step 4	**Pay Now**	Customer clicks Pay Now — system validates the entered details
Step 5	**Invoice Generated**	On success: Payment Invoice is auto-generated with Policy Number, Amount, Date, and Transaction Reference
Step 6	**Policy Activated**	Policy status is set to Active and a confirmation notification is sent to the customer

8.2 End-to-End System Workflow

Step 1	**Authentication**	User registers / logs in → JWT issued → Role authorization applied
Step 2	**Category Setup**	Admin creates property categories and subcategories
Step 3	**Property Registration**	Customer/Agent submits property details → Draft Policy created
Step 4	**Risk Assessment**	Agent/Admin evaluates property value, location risk, construction → Premium calculated
Step 5	**Policy Approval**	Admin reviews → approves/rejects → Policy Number & invoice generated
Step 6	**Payment**	Customer enters amount & card details → clicks Pay Now → Invoice generated → Policy Activated
Step 7	**Claim Submission**	Customer raises claim with damage details and evidence
Step 8	**Claim Review**	Claims Officer verifies → Approved / Rejected
Step 9	**Settlement**	Approved: Settlement = Loss − Deductible → Status: Settled
Step 10	**Rewards**	Referral code used at registration → reward auto-credited on referred user's first activation

8.3 Admin Filter-Based Search — User Management

Administrators can filter the user list using the following criteria. All filters support partial-match queries. Results are paginated (default: 20 per page) and sortable by any column.

Filter Field	Type	Description	Example
Name	Text	Partial match on first name or last name	"John" returns all users with John in name
Email	Text	Partial match on email address	"@gmail" returns all Gmail users
Role	Dropdown	Filter by Admin, Agent, Customer, Claims Officer	Role = Customer
Status	Dropdown	Active / Inactive / Suspended	Status = Active
Date Joined	Range	Filter by registration date range	From: 01-Jan-2024 To: 31-Dec-2024
Referral Code	Text	Filter users who registered with a specific referral code	Code: REF-00123

8.4 Admin Filter-Based Search — Policy Management

Administrators can filter the policy list using the following criteria to efficiently locate, review, and manage insurance policies.

Filter Field	Type	Description	Example
Policy Number	Text	Exact or partial match on policy number	POL-20240001
Customer Name	Text	Partial match on policy holder name	"Sharma"
Policy Status	Dropdown	Draft / Pending Approval / Active / Expired / Cancelled	Status = Active
Property Category	Dropdown	Filter by Residential / Commercial / Industrial / Contents	Category = Residential
Premium Range	Range	Filter by minimum and maximum premium amount	₹5,000 – ₹50,000
Start Date	Range	Filter by policy start date range	01-Jan-2024 to 31-Mar-2024
Expiry Date	Range	Filter by policy expiry date range	Expiring within 30 days
Agent Name	Text	Filter policies handled by a specific agent	"Ramesh"
Claim Status	Dropdown	Policies with active claims: None / Submitted / Settled	Claim Status = Submitted

9. Non-Functional Requirements

9.1 Security

- NFR-01: All endpoints secured with JWT Bearer authentication
- NFR-02: Passwords stored using bcrypt hashing
- NFR-03: RBAC enforced at API controller level
- NFR-04: Input validation on all endpoints to prevent SQL injection and XSS

9.2 Performance

- NFR-05: Pagination and filtering on all list endpoints
- NFR-06: Database queries optimized with appropriate indexes
- NFR-07: API response time ≤ 2 seconds under normal load

9.3 Scalability

- NFR-08: Clean Architecture supports horizontal scaling of backend services

- NFR-09: New insurance categories can be added without core code changes

9.4 Reliability

- NFR-10: Global exception middleware for structured error handling
- NFR-11: Audit trail maintained for all policy, claim, and payment status changes

9.5 Usability

- NFR-12: Angular frontend supports Light and Dark Mode
- NFR-13: Responsive UI across desktop and mobile browsers
- NFR-14: Role-based navigation menus and dashboard analytics

10. System Architecture — Clean Architecture

PIMS is built on a Clean Architecture pattern with four clearly separated layers. Dependencies flow inward: outer layers depend on inner layers; inner layers have no knowledge of outer layers.

API Layer	ASP.NET Core Controllers — exposes RESTful endpoints, handles HTTP requests/responses, JWT middleware, input validation, and routes to Application layer.
Application Layer	Business logic services, use-case orchestration, DTOs, AutoMapper, premium calculation engine, referral reward logic, and claim settlement workflows.
Domain Layer	Core domain entities (Policy, Claim, Property, User, Payment), domain interfaces, business rules, and domain events. No external dependencies.
Infrastructure Layer	EF Core DbContext, repository implementations, SQL Server integration, document storage, email/notification services, and payment gateway adapter.

10.1 API Layer

The API Layer is the entry point of the system. It contains:

- ASP.NET Core Controllers for all HTTP endpoints
- JWT middleware for authentication and authorization
- Request/Response DTOs and AutoMapper profiles
- Global exception handling middleware
- Swagger documentation
- Rate limiting and request logging
  - Endpoints: /api/auth, /api/policies, /api/claims, /api/payments, /api/users, /api/categories

10.2 Application Layer

The Application Layer contains all business logic and orchestrates use-case workflows:

- Service interfaces and implementations (PolicyService, ClaimService, PaymentService, RewardService)

- Premium calculation engine: Premium = Property Value × Risk Factor × Coverage Rate
- Claim settlement engine: Settlement Amount = Loss Amount − Deductible
- Referral reward orchestration — auto-credit on first policy activation
- Filter/search query builders for Admin user and policy management
- Domain event publishing (policy approved, payment succeeded, reward credited)

10.3 Domain Layer

The Domain Layer is the innermost layer with zero external dependencies. It contains:

- Core entities: User, Policy, Claim, Property, Payment, PropertyCategory, PropertySubCategory, RewardTransaction
- Domain interfaces: IRepository<T>, IPolicyRepository, IClaimRepository, IPaymentRepository
- Business rules and invariants (e.g., policy cannot activate without payment)
- Domain events and value objects

10.4 Infrastructure Layer

The Infrastructure Layer provides concrete implementations of domain interfaces:

- EF Core DbContext and repository implementations for SQL Server
- Payment processing adapter implementing IPaymentGateway
- Email/notification service (SMTP or push)
- Document storage service (file system or blob storage)
- Database migrations and seed data

10.5 Frontend Structure (Angular)

Modules: Authentication, Property, Policy, Claims, Admin Dashboard, Reports

- Route Guards to protect role-specific pages
- HTTP Interceptors for automatic JWT token injection
- Filter & Search components wired to Admin User Management and Policy Management pages
- Role-based navigation menus
- Dashboard Analytics with KPIs and charts

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11. Database Design

11.1 Core Tables

Table	Description
Users	User credentials, profile, role assignments, referral code
Roles	System roles: Admin, Agent, Customer, Claims Officer
PropertyCategories	Master property categories

PropertySubCategories	Subcategories linked to master categories
Properties	Registered property records with details and documents
Policies	Insurance policies with status, validity, premium
Claims	Claim records with submission details and status
Payments	Payment records — PaymentId, Amount, Card (last 4 digits), PaymentDate, Status, PolicyId
Documents	Supporting documents for properties and claims
RewardTransactions	Referral reward credits and redemption records

11.2 Database Features

- Primary and Foreign Key constraints for referential integrity
- Indexes on PolicyStatus, ClaimStatus, PaymentStatus, UserId for query performance
- Policy history table for full audit trail
- Claim status timestamps for SLA tracking
- PaymentStatus: Paid / Failed

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12. UML Diagrams

The following UML diagrams describe the structural and behavioural design of the Property Insurance Management System.

12.1 Use Case Diagram

The Use Case Diagram shows interactions between the four system actors (Admin, Agent, Customer, Claims Officer) and all major system use cases including policy management, claim processing, and payment.

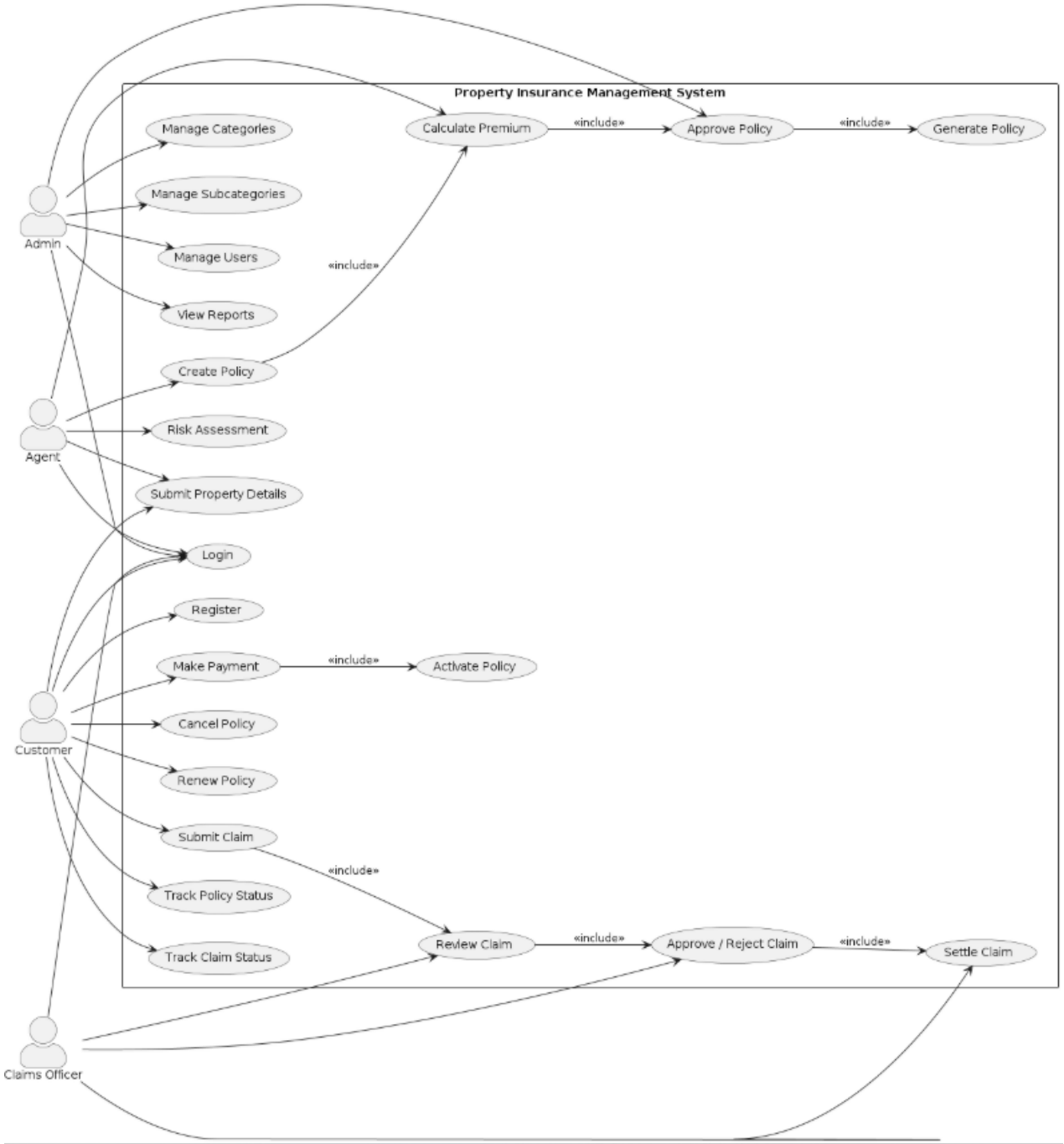

Figure 3 — Use Case Diagram: Property Insurance Management System

12.2 Class Diagram

The Class Diagram illustrates the core domain entities, their attributes, methods, and relationships. Key entities include User, Policy, Property, Claim, Payment, PropertyCategory, and PropertySubCategory.

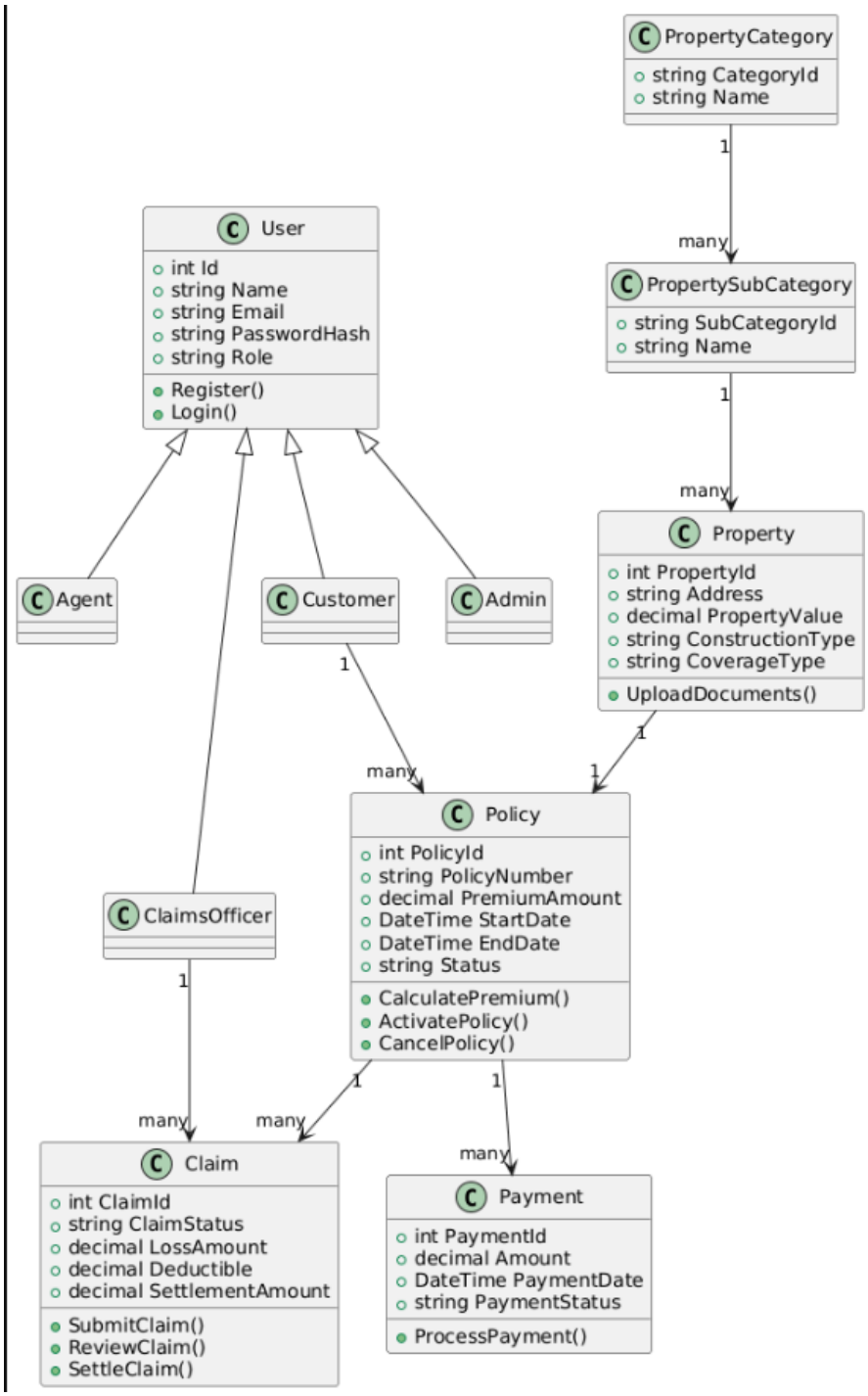

Figure 4 — UML Diagram: Property Insurance Management System

12.3 Activity Diagram

The Activity Diagram captures the complete end-to-end process flow — from user registration and JWT authentication through property submission, premium calculation, policy approval, payment gateway processing, policy activation, claim submission, claim review, and final settlement.

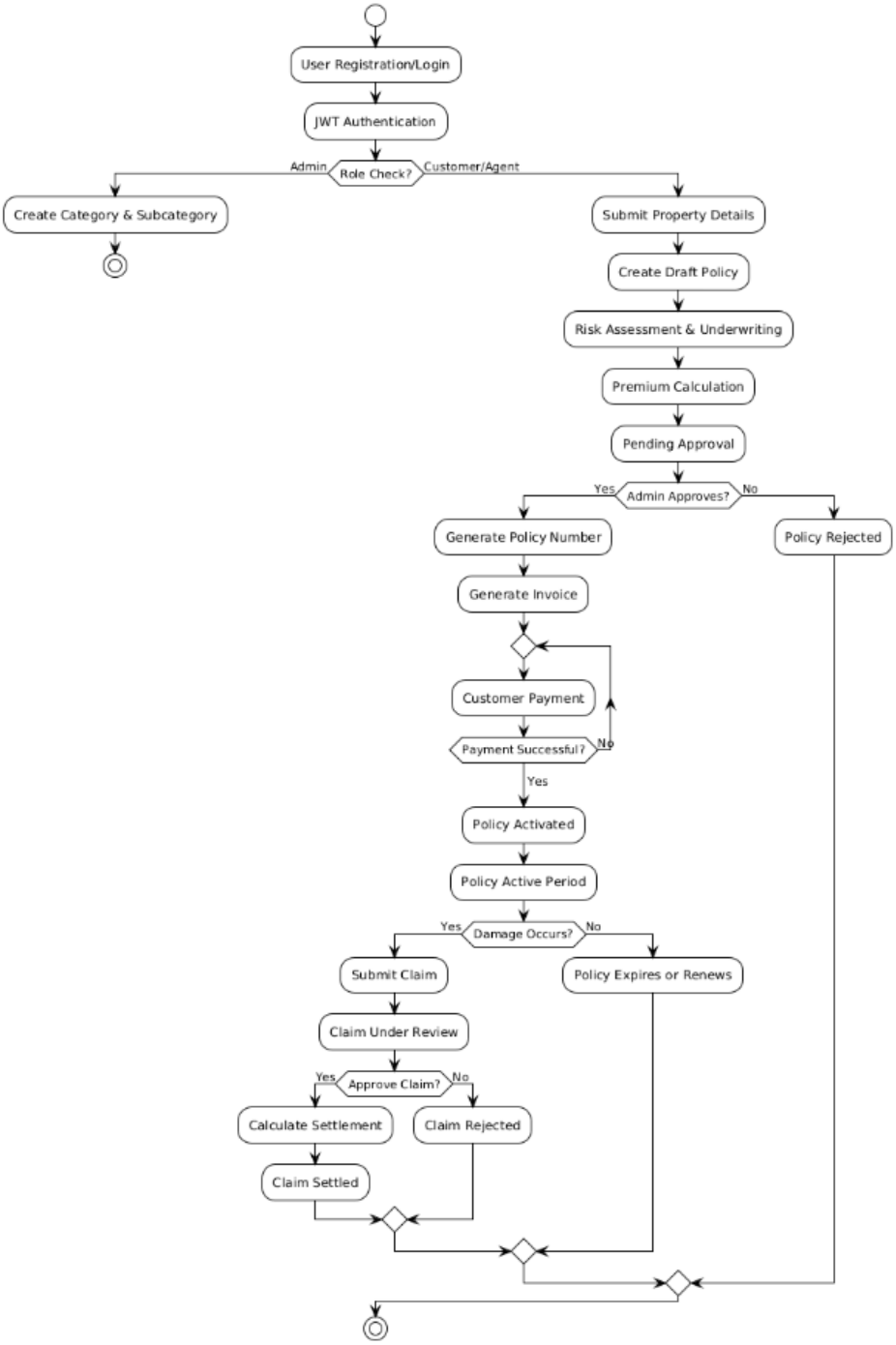

Figure 5 — Activity Diagram: End-to-End Policy & Claim Lifecycle

12.4 Clean Architecture Layer Diagram

The following diagram represents the Clean Architecture dependency flow for PIMS:

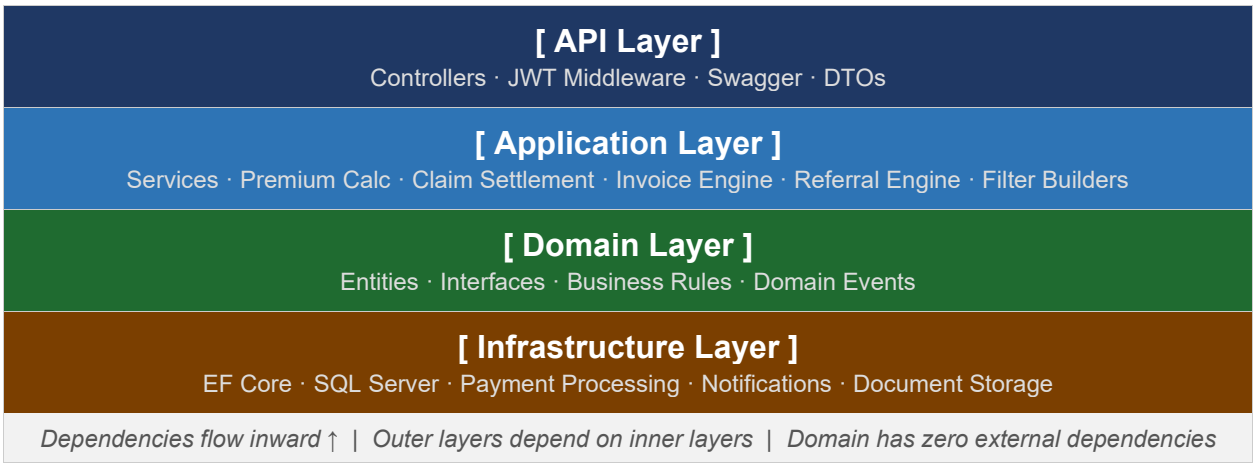

Figure 6 — Clean Architecture Layer Diagram